

Stage life testing with degradation data: a stage-wise hybrid life–degradation test for highly reliable products

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Abstract

In a progressively censored life test, units withdrawn at censoring times are usually meant to be used for other tests, yet standard inferential procedures simply discard them. Stage life testing (SLT) was the first model to bring such units back into the analysis, by re-testing them to failure on a second stage. For highly reliable products this remedy often disappoints: even re-tested units may not fail within a feasible horizon, although their degradation can be measured cheaply and continuously. We therefore propose stage life testing with degradation data (SLT-D). The withdrawn units are moved to an instrumented rig at an elevated stress, where a baseline reading and periodic degradation measurements are taken until the end of the test. Working under an inverse Gaussian degradation process with a power-law time transformation and path-level cumulative exposure, we derive the exact likelihood. It factorizes over conditionally independent blocks because the randomized withdrawal is ignorable, and a variant without the baseline reading is handled by marginalization (an EM scheme). We show, analytically and by profile likelihood, that discarding the withdrawn units leaves only single-stress data, so that extrapolation to the use condition becomes unidentifiable. Failure re-testing is identified but badly behaved: in simulations anchored to stress relaxation of electrical connectors, degradation follow-up reduces the RMSE of the log use-condition 0.1-quantile by factors of 2 to 39 relative to failure re-testing, and it never produces the erratic estimates that re-testing sometimes does. We also derive a planning Fisher information that accounts for mid-test failures of withdrawn units, characterize V-optimal SLT-D plans under a budget, and quantify the price of the stage-wise (sequential) constraint relative to an unconstrained parallel two-level accelerated degradation test. An L_9 array shows that the optimal plan tolerates moderate planning errors, but it also uncovers a collapse mechanism: planning errors that push $F_{x_0}(\tau_1)$ toward 1 empty the withdrawn arm and leave both competing schemes unidentifiable. Plans should therefore keep $F_{x_0}(\tau_1)$ well inside $(0, 1)$. Two refinements build on this. A k -stage version spreads the withdrawal over several change times, improves precision by up to 22% at a matched number of recycled units, and protects the test against the collapse. A Bayesian planning criterion with a pilot-informed prior certifies the plug-in optimum when the pilot is informative and adjusts it only when the pilot is weak. A second case study on GaAs laser diodes, whose data show strong unit-to-unit heterogeneity, marks the limit of these gains: when the life test already sits at the use condition, degradation follow-up no longer improves accuracy under a homogeneous fit, so the value of SLT-D lies in cross-stress extrapolation.

Keywords: accelerated degradation test; progressive censoring; stage life testing; inverse Gaussian process; cumulative exposure; hybrid test plan; V-optimality

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1 Introduction

Highly reliable products leave reliability engineers little choice but to extract information from every unit and every hour of a test [11, 13]. Three strands of literature deal with this economy. The first is *progressive censoring*: living units are removed from a life test at intermediate times to free them for other experiments. It is routinely argued that the withdrawn units are used for further testing [1, 2], yet the information they carry is almost always thrown away in the analysis. Laumen and Cramer [7] closed part of this gap with *stage life testing* (SLT), in which the units withdrawn at a fixed stage-change time are re-tested on a second stage under different conditions and their failure times re-enter the likelihood; Laumen and Cramer [9] extended the model to k stages. The second strand concerns products whose failures are too rare to observe. There, *accelerated degradation tests* (ADT) replace failure times by degradation measurements modelled as stochastic processes, and the inverse Gaussian (IG) process [17, 21] has become a popular choice for monotone degradation. Optimal step-stress ADT (SSADT) plans for the IG process were derived by Wang, Wang, and Duan [16], with later work exploring other optimality criteria [15, 18]; for failure-time step-stress planning under the same censoring scheme see Gouno, Sen, and Balakrishnan [4], and for step-stress inference in general Kundu and Ganguly [5]. The third strand is the *hybrid* accelerated test of Ma et al. [10], which allocates scarce units between an accelerated life test (ALT) run to failure at the highest stress and a constant-stress ADT, and so fuses failure-time and degradation information. The fusion, however, is parallel: different units contribute different data types.

This paper joins the three ideas in a *sequential* hybrid design. The SLT scenario stays as it is: one cohort on a life test, progressive withdrawal at a fixed time τ_1 . What changes is the fate of the withdrawn units. Instead of a failure re-test they receive *degradation monitoring* on an instrumented rig at an elevated stress, so the same physical units contribute failure-time (or censoring) information first and degradation information afterwards. We call the resulting design *stage life testing with degradation data* (SLT-D). It fits situations where the stage-0 life test cannot accommodate measurement (a sealed chamber, a certification test, or a rig without instrumentation) while a bench rig can, and it is most attractive when units are expensive and few, which is the regime that motivates hybrid testing in Ma et al. [10].

Our contributions are as follows.

1. **Model and likelihood.** We formulate SLT-D under an IG degradation process with power-law time transformation and the path-level cumulative-exposure (CE) principle, derive the exact likelihood of all observed data, and prove a block-independence lemma under which the likelihood factorizes into ordinary densities (Section 2–3). A variant in which no baseline reading is taken at withdrawal is handled by one-dimensional marginalization, for which we also describe an EM algorithm (Section 3.3).
2. **Identifiability.** We show that discarding the withdrawn units leaves data from a single stress only, so that use-condition extrapolation becomes unidentifiable (Section 3.4); a profile-likelihood analysis quantifies the damage: inside an approximate 15% likelihood region, the use-condition quantile varies by a factor $e^{13.1}$ under discarding versus $e^{1.1}$ under SLT-D (Figure 7).
3. **Simulation evidence.** In three scenarios anchored to stress relaxation of electrical connectors, degradation follow-up reduces the RMSE of the estimated log use-condition 0.1-quantile by factors of 2–39 relative to SLT failure re-testing at identical test resources, with bounded worst-case behaviour where failure re-testing produces erratic estimates (Section 5). Under $\pm 10\%$ misspecification of all model parameters the hierarchy persists whenever the withdrawn

arm remains populated; we also identify and quantify a design-collapse mechanism whereby planning errors that push $F_{x_0}(\tau_1)$ toward 1 empty the withdrawn arm and both schemes become unidentifiable (Section 5.7).

4. **Planning theory.** We derive the expected Fisher information of the plan, including a refinement that weights each reading by the probability that the withdrawn unit is still operational at that reading; this closes a gap in the standard “all readings” planning approximation. We characterize V-optimal SLT-D plans under a budget (Section 4). For the connector case the price of the stage-wise constraint, relative to an unconstrained parallel two-level ADT, is a factor of about 5.5 in asymptotic variance, and the optimal plan is robust in all nine $L_9(\pm 10\%)$ planning-error scenarios.
5. **Multi-stage withdrawal and Bayesian planning.** We extend SLT-D to k stages with multiple change times; likelihood theory, exact simulation, and a small study show a 22% precision gain at a matched number of recycled units, and protection against the design collapse. We further formulate Bayesian V-optimal planning under a pilot-informed prior: for the connector pilot the criterion certifies the plug-in optimum and departs from it only when the pilot is weak (Sections 2.4, 5.4 and 4.3).

Section 6 fits the IG model to the connector stress-relaxation data, validates our implementation against the published estimates of Ma et al. [10], and adds a second study on GaAs laser data with strong unit heterogeneity. Section 7 concludes. Proofs and derivations are collected in the Appendix.

2 The test scheme and degradation model

2.1 Degradation process and failure

A unit possesses a scalar performance characteristic whose deterioration drives failure. Let $\{Y_x(t), t \geq 0\}$ denote its degradation under standardized stress $x \in [0, 1]$, where $x = 0$ is the use condition and $x = 1$ the highest stress that preserves the failure mechanism, with standardization induced by the physics of the stress (e.g. Arrhenius for temperature; Section 6). We assume an inverse Gaussian process with parameter vector $\theta = (\alpha_0, \alpha_1, \lambda, q)$:

$$Y_x(t) \sim \text{IG}(a_x(t), b_x(t)), \quad a_x(t) = \mu_x \Lambda(t), \quad b_x(t) = \lambda \Lambda(t)^2, \quad \mu_x = e^{\alpha_0 + \alpha_1 x}, \quad \Lambda(t) = t^q, \quad (1)$$

where $\text{IG}(a, b)$ has density $f_{\text{IG}}(y; a, b) = \{b/(2\pi y^3)\}^{1/2} \exp\{-b(y - a)^2/(2a^2 y)\}$, $y > 0$. Increments over disjoint intervals are independent with

$$Y_x(t_2) - Y_x(t_1) \sim \text{IG}(\mu_x \Delta\Lambda, \lambda \Delta\Lambda^2), \quad \Delta\Lambda = t_2^q - t_1^q. \quad (2)$$

The link $\mu_x = e^{\alpha_0 + \alpha_1 x}$ is the standardized form of the Arrhenius/power-law acceleration used by Ma et al. [10]; the power law $\Lambda(t) = t^q$ admits both sub-linear ($q < 1$) and linear degradation. The unit fails when its degradation first crosses a critical level $\omega > 0$. Since IG paths are strictly increasing, the first-passage time $T_x = \inf\{t : Y_x(t) \geq \omega\}$, the law used for variable-stress ALT by Doksum and Høyland [3], has distribution function

$$F_x(t) = P(T_x \leq t) = P(Y_x(t) \geq \omega) = \Phi(u_1) - \exp\left(\frac{2b}{a} + \ln \Phi(-u_2)\right), \quad u_{1,2}(t) = \sqrt{\frac{b(t)}{\omega}} \left(1 \mp \frac{\omega}{a(t)}\right), \quad (3)$$

with a, b from (1); the second term is computed in log space for numerical stability [cf. 16]. The p -quantile $\xi_p(x) = F_x^{-1}(p)$ has the closed-form approximation under the normal approximation of the first passage [10, 20]

$$\xi_p(x) \approx \left[\frac{\mu_x}{4\lambda} \left(z_p + \sqrt{z_p^2 + \frac{4\lambda\omega}{\mu_x^2}} \right)^2 \right]^{1/q}, \quad (4)$$

which we use for delta-method inference (the exact quantile is obtained by bisection on (3)).

2.2 Path-level cumulative exposure

Upon a stress change from x_0 to x_1 at time τ , the future path depends only on the accumulated degradation and the new stress [cf. 16]. A unit that carries level b at the switch has *virtual age* under the new stress

$$w(b) = \left(\frac{b}{\mu_{x_1}} \right)^{1/q}, \quad (5)$$

and its continuation at offset $u > 0$ after the switch is distributed as $Y'_{x_1}(w+u) - Y'_{x_1}(w)$ for a fresh process Y'_{x_1} . Consequently the continuation crosses ω within offset u iff the continuation increment reaches $\omega - b$:

$$G(u | b) = P(\text{cross by } u | b) = S_{IG}(\omega - b; \mu_{x_1}\Delta\Lambda, \lambda\Delta\Lambda^2), \quad \Delta\Lambda = (w+u)^q - w^q, \quad (6)$$

where $S_{IG}(y; a, b) = 1 - F_{IG}(y; a, b)$, with density $g(u | b) = \partial G(u | b) / \partial u$; continuation increments between offsets $u_1 < u_2$ follow (2) with $\Delta\Lambda = (w+u_2)^q - (w+u_1)^q$.

Remark 1 (Two cumulative-exposure notions). *Laumen and Cramer [7] link the stages through a shift of the lifetime distribution, $F_{x_0}(\tau_1) = F_{x_1}(v_1)$ (Sedyakin–Nelson; 12). For process-based models this lifetime shift would require equality of both IG parameters across stages, which generally fails, so the usable analogue is the path-level principle above. The continuation law (6) then plays the role of the shifted stage-1 density in SLT, with the latent level b in place of the deterministic time shift.*

2.3 The SLT-D scheme

n identical units start on stage s_0 (stress x_0) at $t = 0$; failures are recorded. Let $D_1 = \#\{i : T_i \leq \tau_1\}$ with realization d_1 .

1. At the prefixed stage-change time τ_1 , $R_1^* = \lfloor \pi_1(n - d_1) \rfloor$ of the $n - d_1$ survivors (Type-P withdrawal with proportion π_1 ; a Type-M variant with a maximum count is defined analogously, cf. 7) are randomly withdrawn and installed on an *instrumented rig at stress* $x_1 > x_0$. Their degradation is read at the switch (baseline b_k) and at L prefixed offsets $\tau_1 = u_0 < u_1 < \dots < u_L \leq T_{\max}$; readings stop for a unit once a reading reaches ω (failure detected at inspection).
2. The remaining $D_2 = n - d_1 - R_1^*$ units stay on stage s_0 until failure.
3. All observation ceases at the horizon $T_{\max}$; survivors are right-censored.

Figure 1(a)–(b) illustrates a realized $K=1$ world. We compare three uses of the withdrawn units at *identical* $(n, \tau_1, T_{\max})$ and the same withdrawal rule:

- (a) *discard* (progressive censoring with fixed censoring time; PC-FCT, 8): withdrawn units contribute only $S_{x_0}(\tau_1) = 1 - F_{x_0}(\tau_1)$;

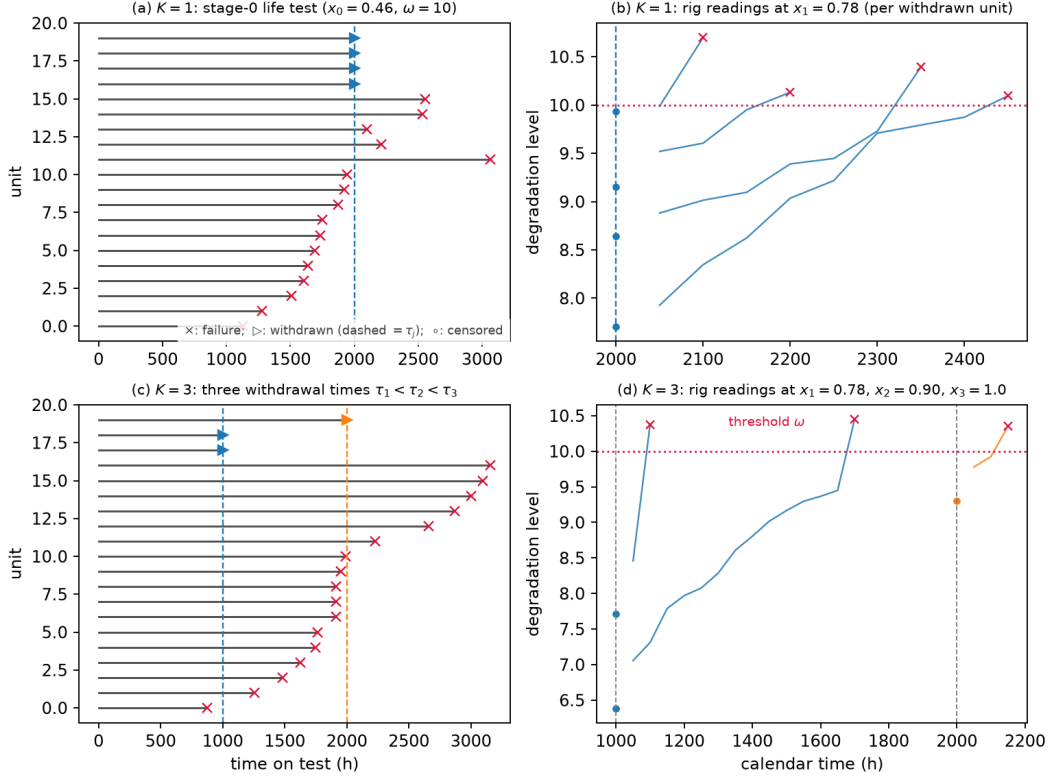


Figure 1: Simulated SLT-D worlds ($n = 20$, $x_0 = 0.46$, $\omega = 10$, $T_{\max} = 5000$ h, readings every 50 h; true θ of Section 6). Panels (a) and (b) show the $K=1$ case: the stage-0 life test, the withdrawal of half the survivors at $\tau_1 = 2000$ h, and the rig readings at $x_1 = 0.78$ (baselines as dots, crossings as $\times$). Panels (c) and (d) show $K=3$: withdrawals at $\tau = (1000, 2000, 3500)$ h with $\pi = (0.15, 0.20, 0.25)$ into rigs at $x = (0.78, 0.90, 1.0)$.

- (b) *SLT* (7): withdrawn units are re-tested at x_1 to failure, censored at $T_{\max}$;
- (c) *SLT-D* (this paper): withdrawn units contribute the baseline reading and the continuation increments.

2.4 Multi-stage extension

The single change point generalizes to K withdrawal times in the obvious way, yielding a k -stage SLT-D that parallels the k -step stage life testing of Laumen and Cramer [9]. Let $\tau_1 < \dots < \tau_K < T_{\max}$ be prefixed change times with withdrawal proportions $\pi_1, \dots, \pi_K$ and rig stresses $x_1, \dots, x_K$ (all exceeding x_0). At τ_j , a randomized subset of $\lfloor \pi_j \cdot (\text{survivors on test}) \rfloor$ units is withdrawn to an instrumented rig at stress x_j , where a baseline reading and the periodic readings are taken as before; the units still on test after τ_K remain on stage s_0 until failure or $T_{\max}$. Randomized withdrawal at *every* stage is ignorable, so Lemma 1 persists verbatim: conditional on the withdrawal history, the stage-0 failure/censoring records and the per-cohort baseline-plus-increment records are independent

with their ordinary laws, and the log-likelihood is

$$\begin{aligned} \ell = & \sum_{i \in \text{failures}} \ln f_{x_0}(t_i) + n_{\text{cens}} \ln \bar{F}_{x_0}(T_{\max}) + \sum_{j=1}^K \sum_{k \in \text{cohort } j} \ln f_{\text{IG}}(b_k; a_{x_0}(\tau_j), b_{x_0}(\tau_j)) \\ & + \sum_{j=1}^K \sum_{k \in \text{cohort } j} \sum_l \ln f_{\text{IG}}(\Delta_{kl}; \mu_{x_j} \Delta \Lambda_{kl}, \lambda \Delta \Lambda_{kl}^2), \end{aligned} \quad (7)$$

with the virtual ages of cohort j computed at stress x_j . Note that no explicit survival term $\ln \bar{F}_{x_0}(\tau_j)$ appears for withdrawn units: the level density at τ_j restricted to $(0, \omega)$ has total mass $\bar{F}_{x_0}(\tau_j)$, so the baseline density itself carries the stage-0 survival information of the withdrawn unit. $K = 1$ recovers Section 2.3; $K \geq 2$ spreads the withdrawal over time, which Section 5.4 shows to hedge against the design-collapse mechanism of Section 5.7 and to improve precision at a matched number of recycled units (Figure 1(c)–(d)).

3 Likelihood and estimation

3.1 Ignorable withdrawal and block independence

Lemma 1. *Conditionally on $D_1 = d_1$, the pre- τ_1 failure times, the post- τ_1 failure/censoring data of the stayers, and the per-unit baseline-plus-increment data of the withdrawn units are independent, with their ordinary (unconditional) laws. In particular, the likelihood is the product of the corresponding densities, and no combinatorial factor depending on θ appears.*

Proof. The withdrawal at τ_1 selects uniformly among survivors, independently of degradation levels; the selection mechanism is therefore ignorable (missing at random), and each unit's record is a draw from its model density on its observation space (failure density f_{x_0} on $(0, \tau_1]$; stayer failure density on $(\tau_1, T_{\max}]$ or point mass $S_{x_0}(T_{\max})$; withdrawn joint density over level and increments on $\{b < \omega\}$). The product form follows; this is the SLT-D analogue of Lemma 3.3 of Laumen and Cramer [7]. $\square$

Writing $\bar{F}_x = 1 - F_x$, the common stage-0 block of all schemes is

$$\ell_{\text{fail}} = \sum_{i \leq d_1} \ln f_{x_0}(t_i) + \left[\sum_{\text{stayers}} \left(\mathbf{1}\{t_i \leq T_{\max}\} \ln f_{x_0}(t_i) + \mathbf{1}\{t_i > T_{\max}\} \ln \bar{F}_{x_0}(T_{\max}) \right) \right], \quad (8)$$

and the withdrawn-arm contributions are

$$(a) \quad \ell_w = r_1^* \ln \bar{F}_{x_0}(\tau_1); \quad (9)$$

$$(b) \quad \ell_w = \sum_{k=1}^{r_1^*} \begin{cases} \ln m_f(u_k), & \text{unit } k \text{ fails at offset } u_k \leq T_{\max} - \tau_1, \\ \ln m_s, & \text{unit } k \text{ censored at } T_{\max}, \end{cases} \quad (10)$$

$$\begin{aligned} m_f(u) &= \int_0^\omega f_{\text{IG}}(b; a_{x_0}(\tau_1), b_{x_0}(\tau_1)) g(u | b) db, \\ m_s &= \int_0^\omega f_{\text{IG}}(b; a_{x_0}(\tau_1), b_{x_0}(\tau_1)) (1 - G(T_{\max} - \tau_1 | b)) db; \end{aligned}$$

$$(c) \quad \ell_w = \sum_{k=1}^{r_1^*} \left[\ln f_{\text{IG}}(b_k; a_{x_0}(\tau_1), b_{x_0}(\tau_1)) + \sum_j \ln f_{\text{IG}}(\Delta_{kj}; \mu_{x_1} \Delta \Lambda_{kj}, \lambda \Delta \Lambda_{kj}^2) \right], \quad (11)$$

where in (11) b_k is the baseline reading, Δ_{kj} the observed increments, $\Delta\Lambda_{kj} = (w_k + u_j)^q - (w_k + u_{j-1})^q$ with $w_k = w(b_k)$ and $u_0 = 0$, and the sum over j extends over the readings taken (including the crossing reading, which is an ordinary IG observation because the level is what is read). Scheme (c) is fully closed-form; the integrals in (10) are one-dimensional and computed by Gauss–Legendre quadrature.

3.2 Estimation and asymptotic inference

The maximum likelihood estimator $\hat{\theta}$ of $\theta = (\alpha_0, \alpha_1, \lambda, q)$ is obtained by multi-start Nelder–Mead search in the unconstrained parameterization $(\alpha_0, \alpha_1, \ln \lambda, \ln q)$, with the covariance from the numerically inverted observed information. Delta-method inference for the use-condition quantile uses the gradient of $\ln \xi_p(0)$ from (4); we report Wald intervals for $\ln \hat{\xi}_p$, whose error is naturally interpreted as a relative error of $\hat{\xi}_p$.

3.3 Variant without baseline reading, and an EM algorithm

If no reading is taken at the switch, the level b_k is latent with prior $f_{\text{IG}}(b; a_{x_0}(\tau_1), b_{x_0}(\tau_1))$ restricted to $b \in (0, \omega)$, and the observed likelihood of unit k ’s readings is

$$L_k = \int_0^\omega f_{\text{IG}}(b) \prod_j f_{\text{IG}}(\Delta_{kj} | b) db, \quad (12)$$

where the first increment is $\ell_{k1} - b$ (level at first reading minus latent baseline) and later increments are observed differences; nodes with $b \geq \ell_{k1}$ contribute zero density because IG paths are monotone, so the integral automatically restricts the latent support. An EM view treats b as missing data: the E-step conditional density is proportional to the integrand of (12) (no closed form; evaluated by quadrature), and the M-step maximizes the expected complete-data log-likelihood numerically (θ enters the increment means through the virtual age $w(b; \theta)$, so no closed-form M-step exists). In our implementation the equivalent direct maximization of the quadrature-approximated marginal (12) is used; simulation shows it remains consistent with a modest precision loss relative to the baseline-reading variant (Section 5).

3.4 Identifiability and the role of the withdrawn units

Proposition 1. *Under scheme (a), all observed data carry the single stress x_0 only. The likelihood then depends on θ solely through the first-passage law F_{x_0} , which identifies at most three functionals of the four parameters; the pair (α_0, α_1) is not separable and $\xi_p(0)$ is not identified.*

The proof is immediate: f_{x_0} , $\bar{F}_{x_0}(\tau_1)$ and $\bar{F}_{x_0}(T_{\max})$ involve θ only through (μ_{x_0}, λ, q) , and $\xi_p(0)$ requires $\mu_0 = e^{\alpha_0}$ and $\mu_{x_0} = e^{\alpha_0 + \alpha_1 x_0}$ separately. Scheme (b) and SLT-D restore identifiability through the stage-1 arm. The practical force of Proposition 1 is illustrated by profile likelihood in Section 5 (Figure 7): with zero stage-0 failures, the approximate 15% profile-likelihood region of scheme (a) contains (α_0, α_1) pairs whose implied $\xi_{0.1}(0)$ spans a factor $e^{13.1}$, versus $e^{1.1}$ for SLT-D on the same data.

4 Planning information and V-optimal designs

4.1 Expected information of an SLT-D plan

For planning we need the Fisher information of the design rather than of a realization. Let $\mathcal{D} = (x_0, \tau_1, \pi_1, x_1, u_1, \dots, u_L)$ denote the plan. Per test unit, the expected information is

$$\begin{aligned} \mathcal{I}(\theta; \mathcal{D}) = & \underbrace{\int_0^{\tau_1} f_{x_0} H_f + (1 - \pi_1) \left[\int_{\tau_1}^{T_{\max}} f_{x_0} H_f + \bar{F}_{x_0}(T_{\max}) H_S \right]}_{\text{stage-0 block}} \\ & + \underbrace{\pi_1 \int_0^{\omega} f_B(b) \left[H_b(b) + \sum_{j=1}^L \mathbf{1}\{\text{alive}_j(b)\} J_j(b)^\top W_j(b) J_j(b) \right]}_{\text{withdrawn block}} db \end{aligned} \quad (13)$$

where $H_f(t) = -\ddot{\ell}_\theta \ln f_{x_0}(t)$ and analogously H_S , H_b for the censoring term and the baseline reading (f_B the level density at τ_1), and for the j -th continuation increment with mean/shape (a_j, s_j) , $W_j = \text{diag}(s_j/a_j^3, 1/2s_j^2)$ is the IG(a, s) information and $J_j = \partial(a_j, s_j)/\partial\theta$ (the second-derivative term vanishes in expectation because the IG score has zero mean; Appendix B). A standard planning approximation evaluates (13) with all L readings. Because IG paths cross ω in finite time, withdrawn units with fast degradation stop contributing readings; we therefore weight the j -th increment by

$$P(\text{alive}_j \mid b) = 1 - G(u_{j-1} \mid b), \quad (14)$$

the probability that the unit is still operational at the previous reading. This refinement is quantitatively important: in our moderate scenario it changes the predicted information by tens of percent (Section 5) and its predictions then match the simulated variances closely (Table 8). The V-optimality criterion is

$$\text{Avar}\{\ln \hat{\xi}_p(0)\} = \nabla_\theta^\top \ln \xi_p(0) (n\mathcal{I})^{-1} \nabla_\theta \ln \xi_p(0), \quad (15)$$

with $\nabla \ln \xi_p$ from (4).

4.2 Constrained and unconstrained optima, and the price of sequentiality

The plan is optimized over $(x_0, \tau_1, \pi_1, x_1, \text{step}, n)$ under a budget

$$n(C_b + C_{\text{mea}} \pi_1 \bar{F}_{x_0}(\tau_1) L) \leq \text{budget}, \quad n \leq n_{\max}, \quad (16)$$

with unit cost C_b and per-reading cost C_{mea} [cf. 10]. As the unconstrained benchmark we optimize a *parallel* two-level constant-stress ADT (n_1 units at x_a , n_2 at x_b , all measured over $(0, T_{\max}]$) under the same budget. The benchmark represents a world in which any unit can be instrumented from the start; the SLT-D optimum represents the constrained world in which the first stage must be a failure-only life test, as in a sealed chamber or a mandated certification run. Their ratio quantifies the *price of the stage-wise constraint*. Table 1 reports both optima for the connector case ($C_b = \$200$, $C_{\text{mea}} = \$5$, budget \$6000, $n_{\max} = 30$, $T_{\max} = 5000$ h): the constrained optimum places stage 0 at the lowest grid stress ($x_0 = 0.10$, about 45°C on the Arrhenius scale), withdraws $\pi_1 = 0.90$ of survivors at $\tau_1 = 1750$ h into a rig at $x_1 = 0.85$ (about 89°C) with readings every 150 h, $n = 20$, achieving $\text{Avar} = 0.036$ (sd = $e^{0.19}$, that is, $\pm 21\%$ on the quantile); the unconstrained parallel ADT concentrates units at $x_a = 0$, the use condition itself, and reaches $\text{Avar} = 0.0066$. The price of

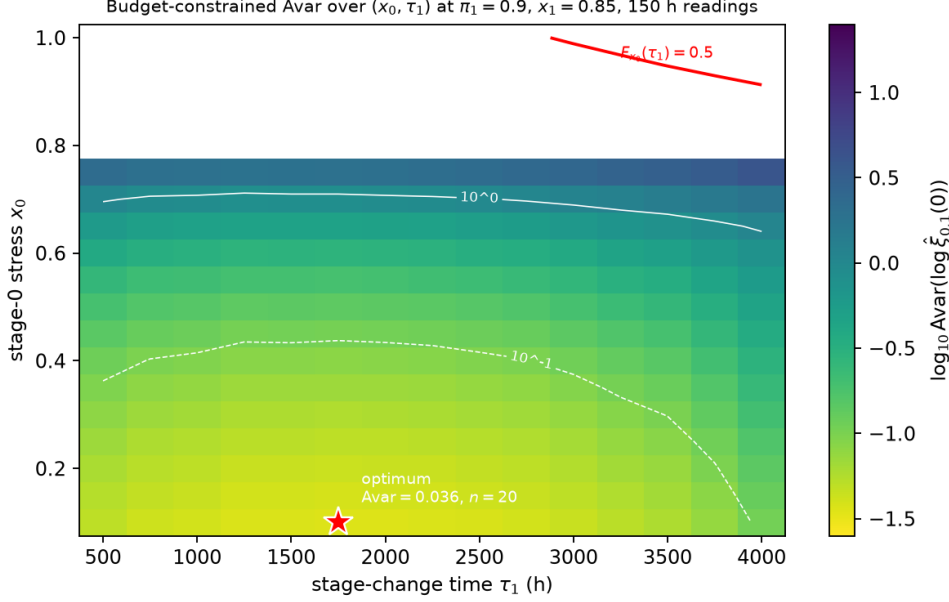


Figure 2: Budget-constrained Avar of $\ln \hat{\xi}_{0.1}(0)$ over (x_0, τ_1) at $\pi_1 = 0.9$, rig stress $x_1 = 0.85$, 150 h readings; n re-derived from the budget at each point (white region: x_0 must lie below the rig stress). The star marks the constrained optimum; the red contour is the survival cliff $F_{x_0}(\tau_1) = 0.5$ of Section 5.7 (the 0.95 contour lies outside this slice).

Table 1: Constrained-optimal SLT-D plan versus the unconstrained parallel two-level ADT at the same budget ($C_b = \$200$ per unit, $C_{mea} = \$5$ per reading, budget \$6,000, $n_{max} = 30$, $T_{max} = 5000$ h; connector parameters; Avar of $\ln \hat{\xi}_{0.1}(0)$).

Design	x_0 / x_a	τ_1	π_1	x_1 / x_b	step	n	Avar
constrained SLT-D	0.10	1750	0.90	0.85	150	20	0.0362
parallel two-level ADT	0.00	—	—	0.20	150	16	0.0066

sequentiality is thus a factor of about 5.5 in asymptotic variance (about 2.3 in standard deviation) for this product and horizon, because the connector’s degradation is measurable at low stress within T_{max} . This echoes the optimal low-stress plans of Ma et al. [10] and Ye et al. [20]. The structure of the optimum is informative in its own right: the criterion pushes the stage-0 stress down and the withdrawal early because, at least for this product, the recycled degradation information matters more than the stage-0 failure information, and a low x_0 maximizes the surviving arm at almost no cost in reading time. Under the sequential constraint the life-test stage acts chiefly as an ageing phase whose surviving units are the real measurement resource. Figure 2 maps the criterion over (x_0, τ_1) at the optimal π_1 , rig stress and cadence: the optimum sits in a broad low-stress basin, far from the survival cliff $F_{x_0}(\tau_1) \rightarrow 1$ (red contour) that marks the collapsed designs of Section 5.7.

When use-condition degradation is too slow to read within the horizon (or when the life test is mandated), the constrained SLT-D plan is the relevant optimum, and Table 2 shows it is robust: under an $L_9(3^4)$ array of $\pm 10\%$ planning errors in $(\alpha_0, \alpha_1, \lambda, q)$, all nine re-optimized plans remain within a factor 1.4 of the base optimum ($Avar \leq 0.052$ at the true parameters), with the qualitative features (low x_0 , large π_1 , 150 h readings) entirely stable.

Table 2: Robustness of the optimal SLT-D plan: $L_9(3^4)$ array of $\pm 10\%$ planning-value errors in $(\alpha_0, \alpha_1, \lambda, q)$; each row re-optimizes the plan at the perturbed planning values and evaluates Avar at the true parameters (base optimum: 0.036).

ϵ_{α_0}	ϵ_{α_1}	ϵ_{λ}	ϵ_q	τ_1	π_1	step	Avar at truth
-10%	-10%	-10%	-10%	2000	0.90	150	0.0391
-10%	+0%	+0%	+0%	2250	0.90	150	0.0370
-10%	+10%	+10%	+10%	3250	0.90	150	0.0518
+0%	-10%	+0%	+10%	2000	0.90	150	0.0364
+0%	+0%	+10%	-10%	1250	0.90	150	0.0369
+0%	+10%	-10%	+0%	2250	0.90	150	0.0370
+10%	-10%	+10%	+0%	1250	0.90	150	0.0369
+10%	+0%	-10%	+10%	2000	0.90	150	0.0364
+10%	+10%	+0%	-10%	1750	0.90	150	0.0362

4.3 Bayesian planning under parameter uncertainty

The L_9 analysis treats planning errors one scenario at a time. An alternative is to average the criterion itself over a prior distribution for θ [cf. 6]. We take the prior to be what a pilot study of the case-study product would supply: a multivariate normal on $z = (\alpha_0, \alpha_1, \ln \lambda, \ln q)$ centered at the pilot MLE with covariance κ times the inverse observed information of the pilot likelihood ($\kappa = 1$ for the 18-unit connector pilot; $\kappa = 4$ emulates a pilot a quarter as informative). The resulting marginal prior standard deviations are 0.19, 0.18, 0.31, and 0.04. Nonlinearity q is pinned down tightly by the within-unit reading trajectories; the stress-acceleration pair and λ remain comparatively uncertain. The Bayesian V-optimal plan minimizes the pre-posterior criterion

$$J(\mathcal{D}) = \mathbb{E}_{\theta \sim \text{prior}} [\text{Avar}\{\ln \hat{\xi}_p(0); \theta, \mathcal{D}\}], \quad (17)$$

evaluated by averaging (15) over 400 prior draws, with the design variables and the budget-feasible n chosen exactly as in Section 4 (the budget is committed at the planning values; only the criterion is averaged).

Table 3 compares the Bayesian-optimal plan with the plug-in plan optimized at the prior mean. At both prior scales the two criteria select the same plan, namely the constrained optimum of Table 1, so the Bayesian criterion serves here as a robustness certificate rather than a correction. Under the full pilot the prior-expected criterion is $J = 0.037$ with 90th percentile 0.041; with a pilot a quarter as informative ($\kappa = 4$) it is $J = 0.038$ with 0.047 (Figure 3b). All of these are close to the plug-in value of 0.036, so the plan’s precision is insensitive to plausible pilot errors. Why averaging leaves the optimum in place can be read off Figure 3(a): at the constrained optimum the stage-0 stress is low ($x_0 = 0.10$) and $F_{x_0}(\tau_1) \approx 0$, the recycled arm is guaranteed for every prior draw, the collapse boundary sits many prior standard deviations away, and the Avar surface is nearly symmetric in the uncertain directions. The expectation is then well approximated by its value at the mean. The Bayesian correction becomes first-order only when the prior mass approaches the survival cliff of Section 5.7, that is, in regimes with $F_{x_0}(\tau_1)$ near 0.5 where collapse draws dominate the expectation and pull the optimum toward earlier withdrawal and lower x_0 ; in those regimes the multi-stage withdrawal of Section 5.4 is the structural safeguard.

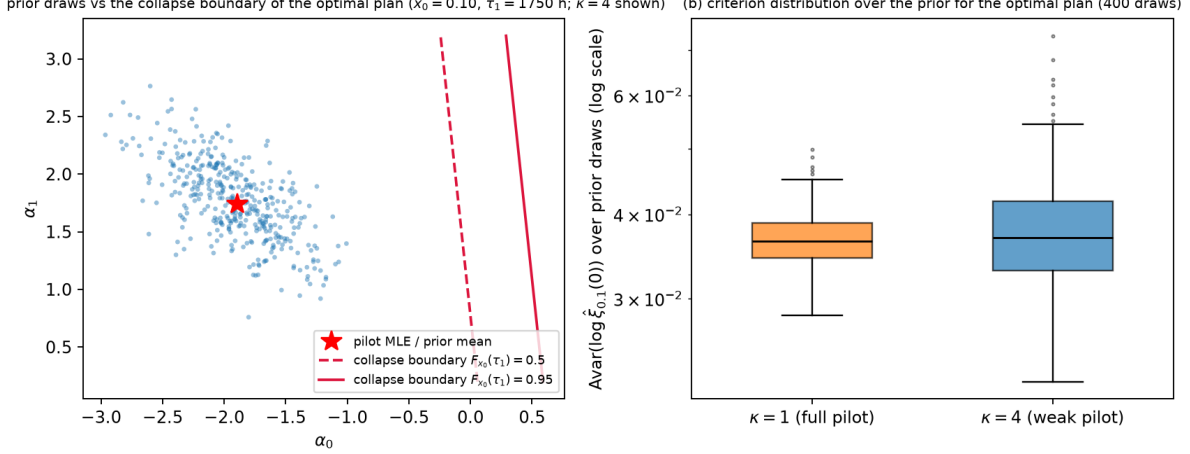


Figure 3: Bayesian planning diagnostics. (a) Prior draws ($\kappa = 4$) in the (α_0, α_1) plane against the collapse boundaries $F_{x_0}(\tau_1) = 0.5/0.95$ of the optimal plan: the prior mass lies far from the cliff, so averaging over the prior cannot move the optimum. (b) Distribution of $\text{Avar}(\ln \hat{\xi}_{0.1}(0))$ over 400 prior draws for the optimal plan under the full ($\kappa = 1$) and a quarter-as-informative ($\kappa = 4$) pilot.

Table 3: Bayesian versus plug-in V-optimal SLT-D plans under pilot-informed priors (400 prior draws; budget and constraints as in Table 1). κ scales the pilot covariance ($\kappa = 1$: the 18-unit pilot). J : prior-expected Avar of $\ln \hat{\xi}_{0.1}(0)$; J_{90} : its 90th percentile over the prior.

Prior	Plan	x_0	τ_1	π_1	x_1	step	n	J	J_{90}	Avar at mean
$\kappa=1$	Bayesian	0.10	1750	0.9	0.85	150	20	0.0368	0.0411	0.0362
$\kappa=1$	plug-in	0.10	1750	0.9	0.85	150	20	0.0368	0.0411	0.0362
$\kappa=4$	Bayesian	0.10	1750	0.9	0.85	150	20	0.0380	0.0472	0.0362
$\kappa=4$	plug-in	0.10	1750	0.9	0.85	150	20	0.0380	0.0472	0.0362

5 Simulation study

5.1 Design of the experiments

True parameters are our fit of the connector data (Section 6), $\theta = (-1.8944, 1.7377, 0.6285, 0.4491)$. Three scenarios (Table 4) span the reliability range: S1 (connector-like: $x_0 = 0.78$ [85°C], $x_1 = 1.0$ [100°C], $\omega = 30$, $n = 20$) with essentially no stage-0 failures; S2 (moderate: $x_0 = 0.46$ [65°C], $x_1 = 0.78$, $\omega = 10$, $n = 20$) with rich failure data; S3 (severe: $x_0 = 0.46$, $x_1 = 0.78$, $\omega = 30$, $n = 30$) with zero failures anywhere on stage 0. All plans use $\tau_1 = 2000$ h, $T_{\max} = 5000$ h, $\pi_1 = 0.5$ (S2 additionally sweeps $\pi_1 \in \{0.25, 0.5, 0.75\}$); readings every 150 h (S1, S3) or 50 h (S2). Each cell aggregates 1000 independent worlds; the three schemes are evaluated on common random numbers. Estimand: $\ln \xi_{0.1}(0)$. Multi-start maximization is initialized from jittered planning values plus one generic start; fits whose numerical information is not positive definite are flagged.

5.2 Information hierarchy

Table 5, Figures 5 and 4 give the headline results; errors are on the log scale, so $\text{RMSE} = 0.5$ means a typical factor- $e^{0.5} \approx 1.6$ error of the quantile. Three findings stand out. First, scheme (a) cannot extrapolate to the use condition in any scenario (RMSE 23 to 54). By Proposition 1 its single-stress

Table 4: Simulation scenarios. Stresses standardized by the Arrhenius rule $x = (1/s_0 - 1/s)/(1/s_0 - 1/s_H)$, $s_0 = 313.15$ K (40°C), $s_H = 373.15$ K (100°C).

	x_0	x_1	ω	n	readings	$F_{x_0}(\tau_1)$
S1 (connector-like)	0.78	1.00	30	20	every 150 h	≈ 0.001
S2 (moderate)	0.46	0.78	10	20	every 50 h	≈ 0.55
S3 (severe)	0.46	0.78	30	30	every 150 h	≈ 0

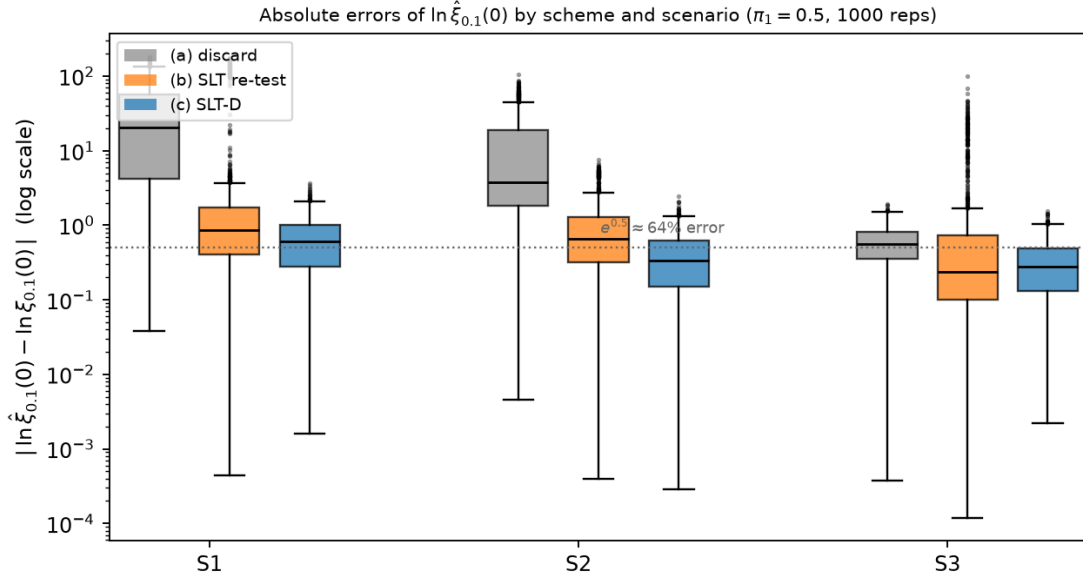


Figure 4: Distributions of the absolute error of $\ln \hat{\xi}_{0.1}(0)$ (log scale; $\pi_1 = 0.5$, 1000 replications). SLT-D concentrates near the $e^{0.5} \approx 64\%$ level or below, whereas scheme (b) develops heavy tails (note the outlier strings in S1/S3) and scheme (a) is off by orders of magnitude.

likelihood leaves (α_0, α_1) unresolved, and the wild point estimates (positive bias up to +33 in log) simply sit on a flat ridge of the likelihood. The apparently tame RMSE of scheme (a) in S3 is an artefact of truth-anchored optimizer starts on that ridge; its numerical information is never positive definite. Second, SLT-D dominates SLT failure re-testing everywhere: RMSE 0.84 against 32.9 in S1 (39 $\times$), 0.53 against 1.82 in S2 (3.4 $\times$), 0.41 against 8.95 in S3 (22 $\times$). The mechanism shows up in the tails and in the inference. The continuation failure times of scheme (b) carry little information about (λ, q) : its 90th-percentile absolute error reaches 11.4 in S3 (versus 0.68 for SLT-D), its observed information is positive definite in only 79 to 93% of fits (versus 99 to 100% for SLT-D), the median standard error in S3 is of order 3×10^4 where the Hessian is near-singular, and the Wald intervals undercover accordingly (78% in S3). The intervals of SLT-D, by contrast, are calibrated: coverage between 88 and 95% across scenarios, with median standard errors of 0.42 to 0.82 that match the empirical RMSE. Third, the gain is largest exactly where accelerated testing matters most. In the failure-free scenarios S1 and S3 the life test produces almost no failures, yet SLT-D still estimates the use-condition quantile to within a factor of about 1.5 ($e^{0.41}$, median error).

Table 5: Information hierarchy at $\pi_1 = 0.5$ (1000 replications). Bias, RMSE, median and 90th percentile of the absolute error of $\ln \hat{\xi}_{0.1}(0)$. Scheme (a): see text regarding S3.

Scenario	Scheme	$\bar{d}_1$	$\bar{r}_1^*$	Bias	RMSE	MdAE	p90AE
S1 (connector-like)	(a)	0.02	9.98	33.435	52.855	20.587	93.500
S1 (connector-like)	(b)	0.02	9.98	9.255	34.409	0.873	4.745
S1 (connector-like)	(c)	0.02	9.98	0.094	0.920	0.609	1.478
S2 (moderate)	(a)	10.60	4.46	11.571	23.283	3.810	44.466
S2 (moderate)	(b)	10.60	4.46	0.438	1.749	0.654	2.855
S2 (moderate)	(c)	10.60	4.46	0.028	0.550	0.335	0.879
S3 (severe)	(a)	0.00	15.00	0.589	0.681	0.562	1.027
S3 (severe)	(b)	0.00	15.00	2.725	8.875	0.241	10.434
S3 (severe)	(c)	0.00	15.00	0.030	0.441	0.279	0.737

Table 6: Effect of the withdrawal proportion π_1 in scenario S2 (1000 replications). Cov.: empirical coverage (%) of the 95% Wald interval.

π_1	Scheme	$\bar{d}_1$	$\bar{r}_1^*$	Bias	RMSE	MdAE	Cov.
0.25	(a)	10.41	2.03	11.617	23.375	3.551	100.0
0.25	(b)	10.41	2.03	0.722	2.085	0.899	94.2
0.25	(c)	10.41	2.03	0.282	1.002	0.425	87.8
0.50	(a)	10.60	4.46	11.571	23.283	3.810	100.0
0.50	(b)	10.60	4.46	0.438	1.749	0.654	92.0
0.50	(c)	10.60	4.46	0.028	0.550	0.335	92.6
0.75	(a)	10.39	6.83	12.666	24.809	3.735	99.9
0.75	(b)	10.39	6.83	0.522	1.874	0.645	94.1
0.75	(c)	10.39	6.83	-0.005	0.471	0.305	95.3

5.3 Withdrawal fraction

Table 6 sweeps π_1 in S2. SLT-D improves monotonically in π_1 (RMSE $0.95 \rightarrow 0.53 \rightarrow 0.46$ for $\pi_1 = 0.25, 0.5, 0.75$): within the fixed budget of units, the more survivors are recycled into measurement, the better, as long as stage 0 retains its (censoring) information. Scheme (b) improves similarly but remains 2–3.8 $\times$ worse at each π_1 ; even with only $\pi_1 = 0.25$ withdrawn, SLT-D is at least as good as scheme (b) at $\pi_1 = 0.75$. Figure 6 visualizes the sweep.

5.4 Multi-stage withdrawal

Table 7 compares $K = 1, 2, 3$ SLT-D plans in scenario S2 with $n = 20$ units, $T_{\max} = 5000$ h, readings every 50 h, and expected numbers of withdrawn units matched at 3.9–4.5: $K=1$ withdraws half the survivors at $\tau_1 = 2000$ h into a $x_1 = 0.78$ rig; $K=2$ adds a second change at $\tau_2 = 3000$ h into a $x_2 = 1.0$ rig ($\pi = (0.30, 0.30)$); $K=3$ spreads withdrawal as $(\tau, \pi, x) = (1000, 0.15, 0.78), (2000, 0.20, 0.90), (3500, 0.25, 1.00)$. Spreading the withdrawal improves precision: the RMSE of $\ln \hat{\xi}_{0.1}(0)$ falls from 0.54 ($K=1$) to 0.46 ($K=2$) and 0.42 ($K=3$), a 22% reduction. Two effects drive this. Earlier cohorts are withdrawn while $\bar{F}_{x_0}(\tau) \approx 1$ units remain, so the recycled arm stays large; in the language of Section 5.7, the plan moves away from the $F_{x_0}(\tau_1)$ cliff (none of the $K \geq 2$ replications collapsed, and the design is protected against planning errors that would empty a single late cohort). In addition, the cohorts observe degradation at two or three distinct

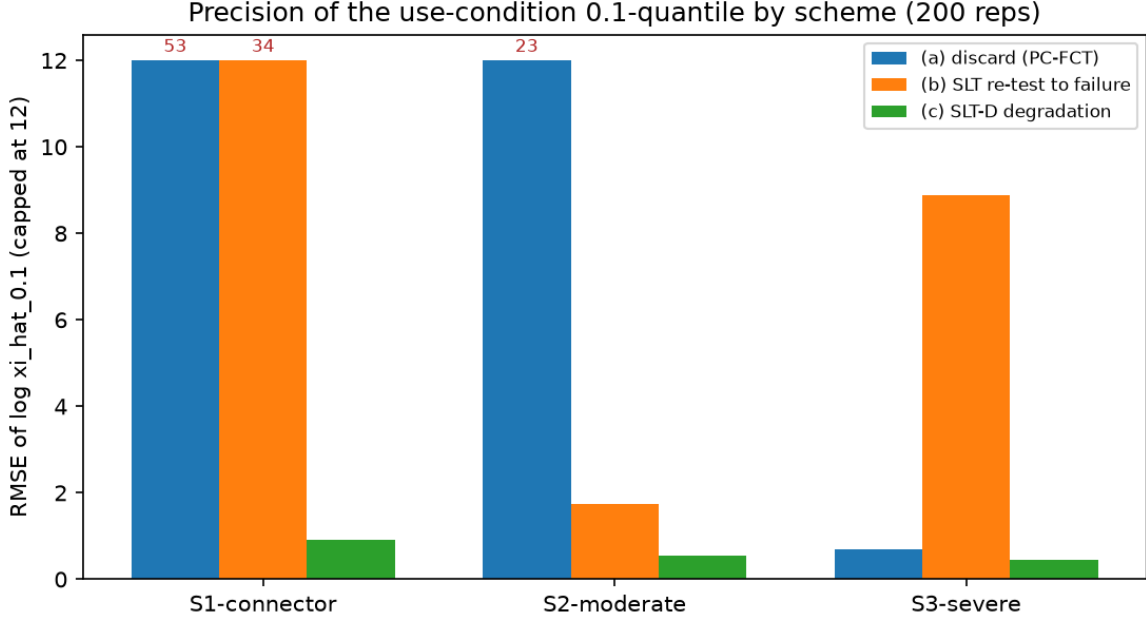


Figure 5: RMSE of $\ln \hat{\xi}_{0.1}(0)$ by scheme and scenario (bars capped at 12 for display; printed values above capped bars). SLT-D (scheme c) dominates throughout.

Table 7: Multi-stage SLT-D in scenario S2 (500 replications): plans matched in units ($n = 20$) and expected withdrawals; $\ln \hat{\xi}_{0.1}(0)$ errors; r = number of readings per replication.

K	stages (τ, π, x)	$\mathbb{E}[R_1^*]$	$\mathbb{E}[\#\text{readings}]$	Bias	RMSE	MdAE	$\text{RMSE} \cdot \sqrt{r}$
1	(2000, 0.50, 0.78)	4.49	14.7	+0.045	0.540	0.337	2.07
2	(1500, 0.30, 0.78); (3000, 0.30, 1)	4.40	19.8	+0.021	0.458	0.302	2.04
3	(1000, 0.15, 0.78); (2000, 0.20, 0.90); (3500, 0.25, 1)	3.87	20.6	-0.001	0.421	0.308	1.91

stresses, which separates the acceleration parameters more cleanly. The gain has a price: with the reading cadence fixed, later-drawn cohorts remain under measurement longer and the expected total number of readings rises from 14.7 to about 20. Normalized per reading, however, efficiency still improves ($\text{RMSE} \times \sqrt{\text{readings}}$: $2.07 \rightarrow 2.04 \rightarrow 1.91$), so multi-stage withdrawal is the sensible default whenever rig time is cheaper than units.

5.5 Profile-likelihood illustration of unidentifiability

Figure 7 profiles the log-likelihood over (α_0, α_1) (maximized over (λ, q)) on a common S3 world with zero stage-0 failures. Scheme (a) exhibits a ridge along $\mu_{x_0} = e^{\alpha_0 + \alpha_1 x_0} = \text{const}$, exactly as Proposition 1 predicts. Within the approximate 15% region the implied $\ln \xi_{0.1}(0)$ ranges over $[4.7, 17.8]$, a factor $e^{13.1}$; SLT-D concentrates around the truth with a range of $e^{1.1}$.

5.6 Planning formula versus simulated variance

Table 8 compares the planning asymptotic variance (15) (with the alive-weighting (14)) against the empirical variance of $\ln \hat{\xi}_{0.1}(0)$ across the 1000 replications of SLT-D. Agreement is close throughout (S1: 0.66 vs 0.84; S2: 0.19 vs 0.30; S3: 0.18 vs 0.19), supporting the use of (13)–(15) for

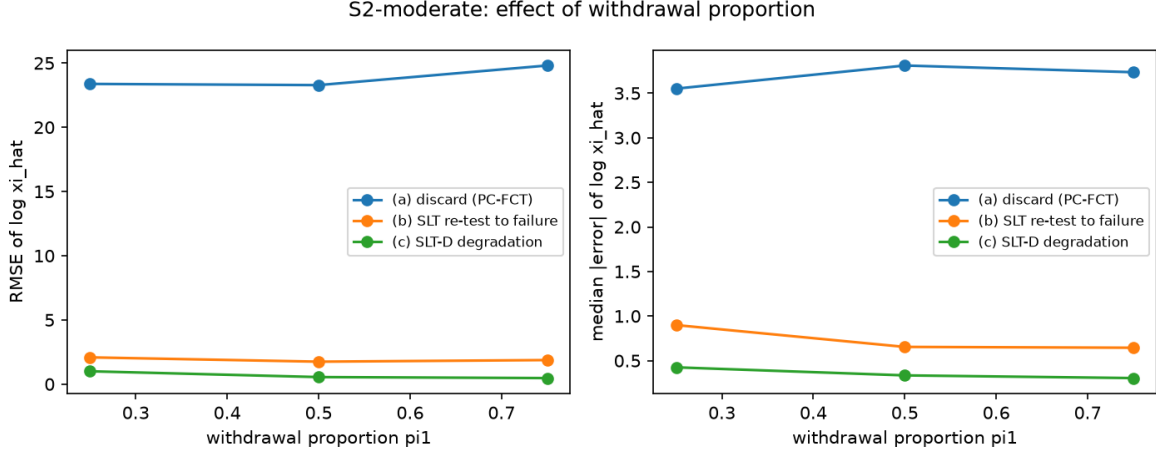


Figure 6: Effect of the withdrawal proportion π_1 in scenario S2 (schemes (b) and (c); 1000 replications).

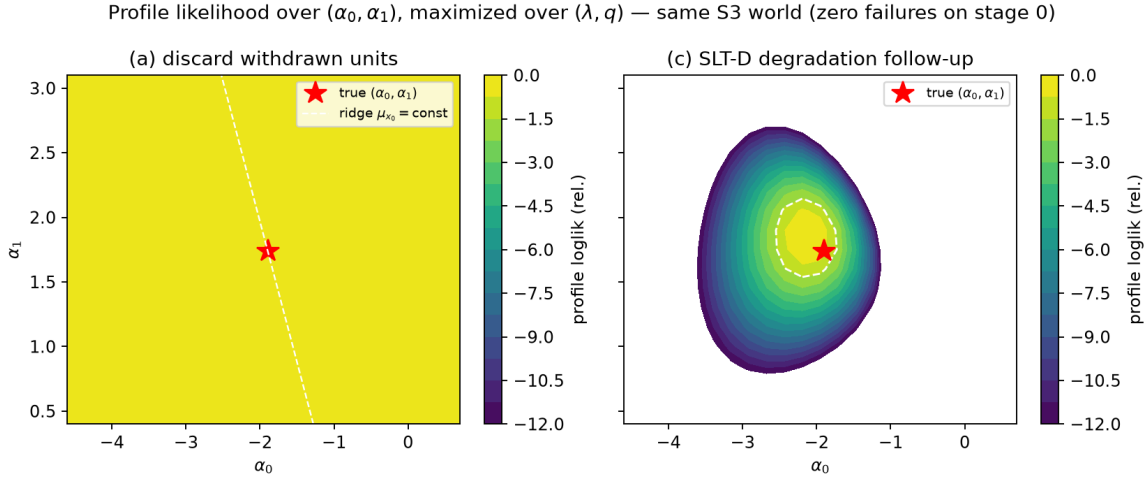


Figure 7: Profile log-likelihood over (α_0, α_1) , maximized over (λ, q) , for scheme (a) (left) and SLT-D (right) on the same realization with zero stage-0 failures (S3). White contour: approximate 15% region; the dashed line in the left panel is the $\mu_{x_0} = \text{const}$ ridge.

design optimization. Without the alive-weighting the planning formula overstates the information substantially whenever withdrawn units fail mid-test (in S2 by about a factor 2.1 in the dominant eigenvalue; Section 4).

5.7 Robustness of the hierarchy to parameter misspecification

All estimators so far use the true θ to initialize and all designs were constructed at the true values. Table 9 repeats the S2 comparison with the *true* parameters perturbed in an $L_9(3^4)$ array of $\pm 10\%$ errors on $(\alpha_0, \alpha_1, \lambda, q)$ (the analysis still being initialized at the unperturbed values, as when planning values come from a pilot study). The expected number of withdrawn units under the perturbed truth, $\mathbb{E}[R_1^*] = \sum_k \binom{20}{k} F_{x_0}(\tau_1)^k \bar{F}_{x_0}(\tau_1)^{20-k} \lfloor (20-k)/2 \rfloor$, is shown to expose the mechanism. Three regimes appear. (i) A healthy arm (combinations 1, 5, 6, 7, 9): the hierarchy holds, with SLT-D better in RMSE by factors of 1.8 to 42. The largest factor arises in combination 9, where scheme (b)'s RMSE is dominated by a catastrophic tail (MdAE 0.32 against RMSE 15.5), exactly as at the nominal

Table 8: Predicted planning Avar of $\ln \hat{\xi}_{0.1}(0)$ versus empirical variance across 1000 replications (SLT-D, $\pi_1 = 0.5$).

Design	n	predicted Avar($\log \hat{\xi}_{0.1}$)	empirical var
S1	20	0.658	0.839
S2	20	0.192	0.303
S3	30	0.175	0.194

Table 9: Estimation robustness: $L_9(3^4)$ array of $\pm 10\%$ errors in the true parameters (S2, 300 replications per combination; errors of $\ln \hat{\xi}_{0.1}(0)$).

Combo	$\mathbb{E}[r_1^*]$	scheme (b)			scheme (c)		
		Bias	RMSE	MdAE	Bias	RMSE	MdAE
1	8.8	0.132	1.205	0.608	-0.038	0.420	0.267
2	0.6	3.401	11.863	1.964	3.448	11.827	1.803
3	0.0	13.127	26.074	3.557	13.127	26.074	3.557
4	0.0	10.961	20.139	3.803	10.989	20.144	3.811
5	9.8	0.096	0.651	0.379	0.004	0.339	0.217
6	2.5	0.574	2.091	0.883	0.322	0.923	0.535
7	9.8	0.053	0.497	0.324	-0.016	0.274	0.187
8	0.4	5.350	13.247	2.242	5.453	13.221	2.138
9	10.0	2.739	15.519	0.324	0.040	0.371	0.247

parameters. (ii) A collapsed arm (combinations 3 and 4): the perturbed parameters push μ_{x_0} up so far that $F_{x_0}(\tau_1) \approx 1$; essentially nothing survives to τ_1 , both schemes receive identical stage-0-only data, and both inherit the unidentifiability of Proposition 1 (identical RMSE of about 20 to 26). (iii) A thin arm (combinations 2 and 8): $\mathbb{E}[R_1^*] < 1$; the stage-0 block again dominates and both schemes degrade, though scheme (c)’s median error stays moderate (MdAE 1.8 to 2.1). The lesson here is more about design than about estimation. Near the knee $a_{x_0}(\tau_1) \approx \omega$ the quantity $F_{x_0}(\tau_1)$ reacts very sensitively to μ_{x_0} , and the S2 plan sits right there ($a_{x_0}(\tau_1) \approx 8.7$ against $\omega = 10$), so a planning error of $\pm 10\%$ in (α_0, α_1) swings $F_{x_0}(\tau_1)$ across most of $(0, 1)$. A robust SLT-D plan should therefore place $F_{x_0}(\tau_1)$ well inside $(0, 1)$, say between 0.3 and 0.8, rather than at the cliff edge; the plan-level L_9 analysis of Section 4 points the same way, and we return to this point in Section 7.

6 Case studies

6.1 Stress relaxation of electrical connectors

We use the stress-relaxation data of a type of electrical connector, presented as Example 8.7 of Yang [19] and tabulated by Wang, Wang, and Duan [16] and Ma et al. [10]: Figure 8 shows the 18 observed paths together with the fitted IG mean degradation $\hat{\mu}_x t^{\hat{q}}$ at each temperature. The data comprise six units each at 65, 85 and 100°C, with 10–11 readings per unit over up to 2810 h; failure is declared at stress relaxation $\omega = 30\%$; the Arrhenius standardization gives $x(65^\circ\text{C}) = 0.460$, $x(85^\circ\text{C}) = 0.781$, $x(100^\circ\text{C}) = 1$. Fitting (1)–(2) by maximum likelihood yields $\hat{\theta} = (-1.894, 1.738, 0.629, 0.449)$ with $\hat{\xi}_{0.1}(40^\circ\text{C}) = 1.20 \times 10^5$ h, in close agreement with the published estimates $(-1.88, 1.73, 0.653, 0.449)$ and 1.16×10^5 h of Ma et al. [10]. This agreement validates our likelihood implementation; the remaining differences reflect rounding in the published table.

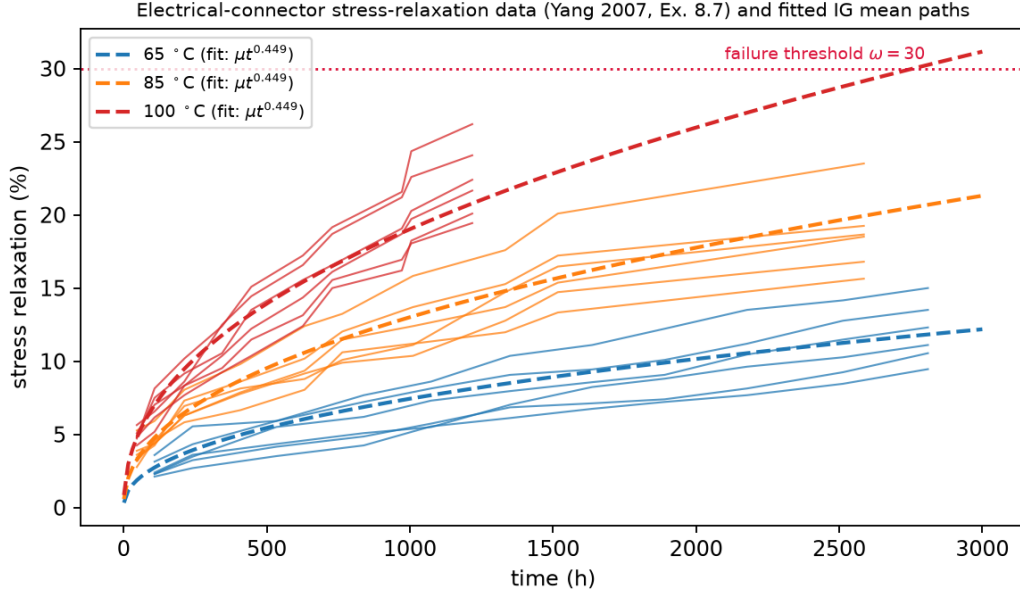


Figure 8: Stress-relaxation paths of the 18 connector units (thin lines) and the fitted IG mean degradation $\hat{\mu}_x t^{\hat{q}}$ (dashed) at the three test temperatures; the dotted line is the failure threshold $\omega = 30\%$.

Re-using these values as ground truth anchors the simulation study (Section 5) and the planning analysis (Section 4). For this product, an SLT-D with 20 units (a life test at $x_0 \approx 45^\circ\text{C}$ for 1750 h, then 90% of survivors withdrawn into an $\approx 89^\circ\text{C}$ rig read every 150 h until 5000 h) estimates the use-condition 0.1-quantile with a planned relative precision of about $\pm 21\%$, and not a single failure needs to occur.

6.2 A second product: GaAs laser diodes under unit heterogeneity

The connector analysis assumed a homogeneous population; real degradation data often violate this, and the GaAs laser dataset of Meeker and Escobar [11] (p. 339) is a canonical example. The degradation is the increase of the operating current needed to maintain constant light output, read every 250 h up to 4000 h for 15 lasers, with failure declared at a 10% increase ($\omega = 10$); three units are observed to cross the threshold (Figure 9). The paths are nearly linear (the homogeneous fit gives $\hat{q} = 1.00$ with $\hat{\mu} = 0.0021$) but strongly heterogeneous across units, with a per-unit rate CV of about 0.22. This is why Wang and Xu [17] and Peng and Tseng [14] introduced random-effect extensions of the IG process for these data.

The data allow us to stress the design comparison in a deliberately misspecified setting. The simulation truth carries the measured heterogeneity (rates μ_i lognormal with log standard deviation 0.21 around the fitted mean $\bar{\mu} = 0.0021$; $q = 1$; within-unit dispersion $\lambda = 3 \times 10^{-5}$), while the estimators remain the homogeneous MLEs that a practitioner without a random-effect model would fit. The estimand is the *population* 0.1-quantile of the failure time, $\xi_{0.1}(0) = 3611$ h. Three designs at the historical resources ($n = 15$, $T_{\max} = 4000$ h, 250 h cadence) are compared in Table 10: the historical full monitoring of all 15 units from $t = 0$; an SLT-D that leaves the life test uninstrumented, withdraws 50% of survivors at $\tau_1 = 2000$ h into a rig with Arrhenius acceleration ($E_a = 0.7$ eV between 300 K and 350 K, $\text{AF} \approx 48$); and the pure life test.

The results trace the limits of SLT-D (Table 10). All three designs overestimate $\xi_{0.1}$, by 0.04

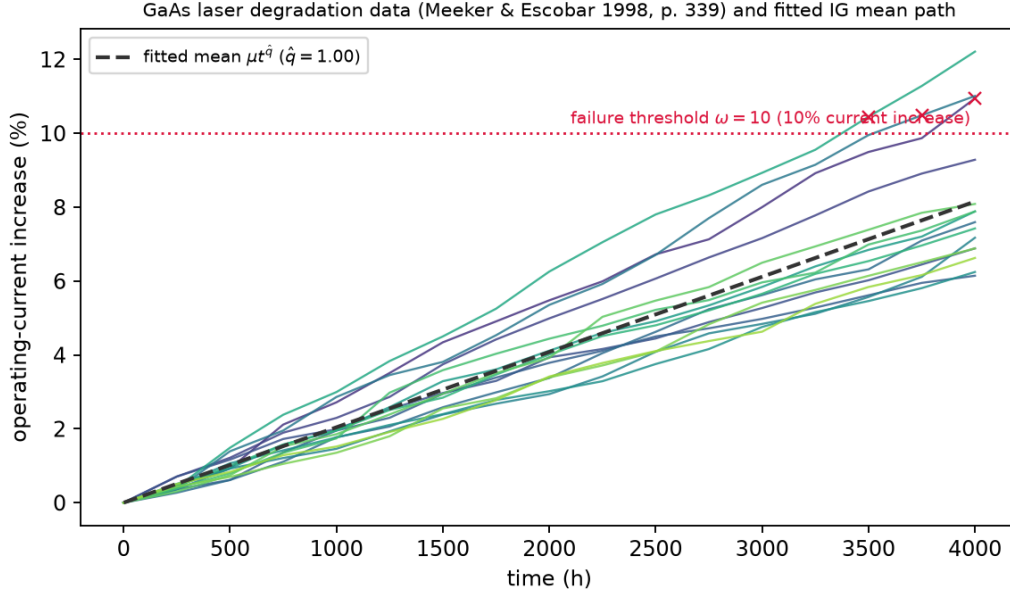


Figure 9: Operating-current increase of the 15 GaAs lasers (Meeker and Escobar [1998], p. 339), the fitted homogeneous IG mean path ($\hat{q} = 1.00$), and the failure threshold $\omega = 10$; crosses mark the three observed crossings. The spread of the paths at 4000 h (6.1–12.2) visualizes the unit heterogeneity.

to 0.14 in log terms; the homogeneous likelihood cannot represent the mixture, and the observed 3/15 crossings, set against a fitted $F(4000\text{ h}) \approx 0.02$, would flag this immediately in practice. The ranking of typical accuracy (median error) is more surprising: the pure life test does best (0.068), SLT-D comes second (0.099), and full degradation monitoring does worst (0.141). There is a simple explanation. The failure/censoring likelihood observes the marginal crossing law, with the heterogeneity already integrated out. The homogeneous increment likelihood instead fits pooled increments, which no homogeneous model actually generates, so the additional degradation information does not translate into better quantile estimates. On robustness, neither full monitoring nor SLT-D produced a degenerate fit in 500 replications, while the life test did once, and the typical error of SLT-D matches that of full monitoring even though fewer than half of the units are instrumented.

Taken together, the two case studies indicate where SLT-D pays off. The gains concentrate in applications where the first stage runs off the use condition, so that discarding the withdrawn units is unidentifiable and cross-stress extrapolation is the central inferential problem, and where unit heterogeneity is mild. When the life test already sits at the use condition and the units are strongly heterogeneous, the failure/censoring data of the life test alone are competitive, and random-effect models [cf. 14] are the better investment.

7 Conclusions

Stage life testing posed the question of what the withdrawn units of a progressively censored life test are worth. Our answer is that, used as degradation-monitoring units, they are worth a great deal. The SLT-D likelihood is exact and available in closed form, randomized withdrawal makes every block of the data conditionally independent, and discarding the withdrawn units destroys the stress transfer needed for extrapolation. Failure re-testing restores identifiability but behaves

Table 10: Laser case study: design comparison under data-calibrated heterogeneity (500 replications; $n = 15$, $T_{\max} = 4000$ h, 250 h cadence; errors of $\ln \hat{\xi}_{0.1}(0)$ against the population quantile 3611 h).

Design	monitored units	degenerate fits	Bias	RMSE	MdAE	p_{90} AE
full monitoring (historical)	15 from $t=0$	0/500	+0.133	0.156	0.141	0.239
SLT-D ($\tau_1=2000$, $\pi_1=0.5$)	≈ 7 from τ_1	0/500	+0.139	0.230	0.099	0.358
life test only	0	1/500	+0.042	0.193	0.068	0.216

erratically, while degradation follow-up was uniformly superior in our simulations. The planning information, refined to account for mid-test failures of withdrawn units, predicts the simulated precision well and yields V-optimal plans. Two refinements strengthen the design further. Spreading the withdrawal over several change times improves precision at a matched number of recycled units and protects the plan against the survival cliff of a single change time. Bayesian planning under a pilot-informed prior turns planning-parameter uncertainty into an explicit criterion; for an informative pilot it certifies the plug-in optimum, and for a weak pilot it fine-tunes that optimum. The design comparison also states the limits of the proposal. When any unit can be instrumented from the start and use-condition degradation is readable within the horizon, a parallel low-stress ADT remains preferable; in the connector case the sequential constraint costs a factor of about 5.5 in asymptotic variance. SLT-D is the design of choice when the first stage must be a failure-only test, as in sealed chambers or mandated certification runs, or when scarce units have to serve both purposes. The laser study adds a boundary condition: when the life test already sits at the use condition and the units are strongly heterogeneous, a homogeneous SLT-D fit inherits the misspecification bias, the pure life test becomes competitive, and random-effect models are the more productive route.

Several extensions are worth pursuing: optimal design with Type-M withdrawal and adaptive change times; Bayesian planning with hierarchical priors elicited from several historical tests; and random effects across units [cf. 14], which would break the clean factorization of Lemma 1 and call for the EM machinery of Section 3.3 at the design stage as well.

Reproducibility

All computations are implemented in Python/NumPy/SciPy: the likelihoods, simulators validated by score identities against the generating model, quadrature checks against Monte Carlo, the 15,000-fit simulation study, and the design optimization. Source code, generated tables, and figures accompany this manuscript. The GaAs laser data were obtained from the public repository of Meeker’s repeated-measured degradation datasets (file GaAsLaser.csv).

A First-passage and continuation laws

Equation (3) follows from $P(T_x \leq t) = P(Y_x(t) \geq \omega)$ and the closed-form IG survival function $S_{IG}(y; a, b) = \Phi(\sqrt{b/y}(1 - y/a)) - e^{2b/a}\Phi(-\sqrt{b/y}(1 + y/a))$; monotonicity of IG paths makes the hitting event equivalent to the level event, and inversion sampling of T_x is exact. For the continuation, the path-level CE principle gives virtual age (5); the crossing event by offset u is $\{Y'_{x_1}(w + u) \geq \omega\}$, i.e. the continuation increment over $(w, w + u]$ reaching $\omega - b$, which is (6); differentiating in u yields $g(u | b)$ (computed by central differences).

B Expected information

For a single observation $y \sim \text{IG}(a, b)$ with (a, b) functions of θ , the expected information is $J^\top W J$ with $W = \text{diag}(b/a^3, 1/2b^2)$ and $J = \partial(a, b)/\partial\theta$: the second-derivative contribution of $\partial J/\partial\theta$ vanishes in expectation because the IG score has zero mean, and $\mathbb{E}[-\partial^2 \ln f/\partial(a, b)^2] = \text{diag}(b/a^3, 1/2b^2)$ (verified numerically to 10^{-2} relative error). Summing over the record types of Lemma 1 with their probabilities gives (13); the alive-weighting (14) follows because reading j is observed iff the unit has not crossed ω by u_{j-1} , an event of probability $1 - G(u_{j-1} | b)$ conditionally on the latent (or observed) level b , and the crossing reading itself remains an ordinary IG observation.

C EM for the variant without baseline reading

With complete data $(b, \Delta_1, \dots, \Delta_L)$ for a withdrawn unit, the complete-data log-likelihood is the sum of the level log-density at τ_1 and the increment log-densities of Section 3. The E-step evaluates, by quadrature, the conditional density of b given the observed increments (proportional to the integrand of (12) on $(0, \min(\ell_1, \omega))$), together with any conditional moments that are needed; the M-step maximizes the expected complete-data log-likelihood numerically. Because the increment means depend on θ through the virtual age $w(b; \theta)$, the M-step has no closed form, and direct maximization of the marginal likelihood (12) (used here) is the equivalent one-step alternative; the observed-data information follows from the numerical Hessian of (12) (Louis-type corrections are unnecessary at the accuracy required for Wald intervals).

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